

Day 10: Adding Categorical Predictors

Recap: `bikes` MLR

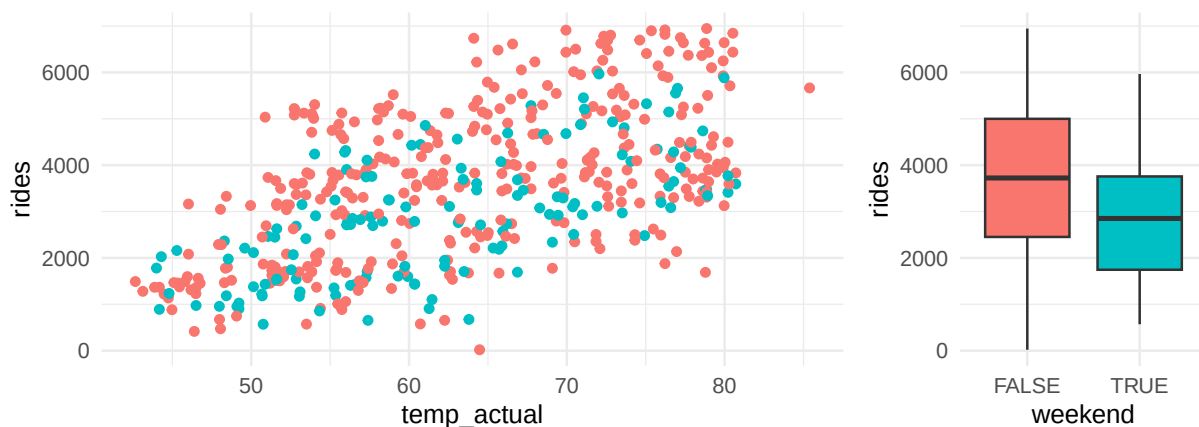
Last time we added a second quantitative predictor, `windspeed`, to our model for `rides`:

$$\widehat{\text{rides}} = -1525.587.02 \cdot \text{temp_actual} - 38.63 \cdot \text{windspeed}$$

term	estimate	std. error	t	p-value
Intercept	-1525.538	398.281	-3.83	<0.001
temp_actual	87.018	5.629	15.46	<0.001
windspeed	-38.628	10.355	-3.73	<0.001

$$R^2 = 0.35$$

Today's plan: drop `windspeed`, add `weekend`



A new example: lung function and smoking

Goal: investigate the association between smoking and lung capacity using data from 345 adolescents ages 10-19.

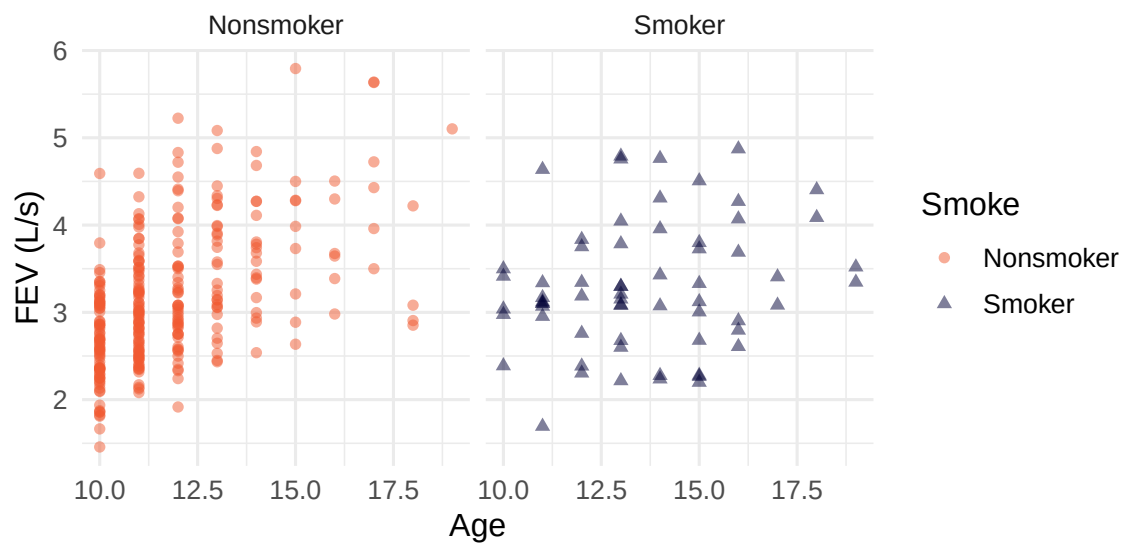
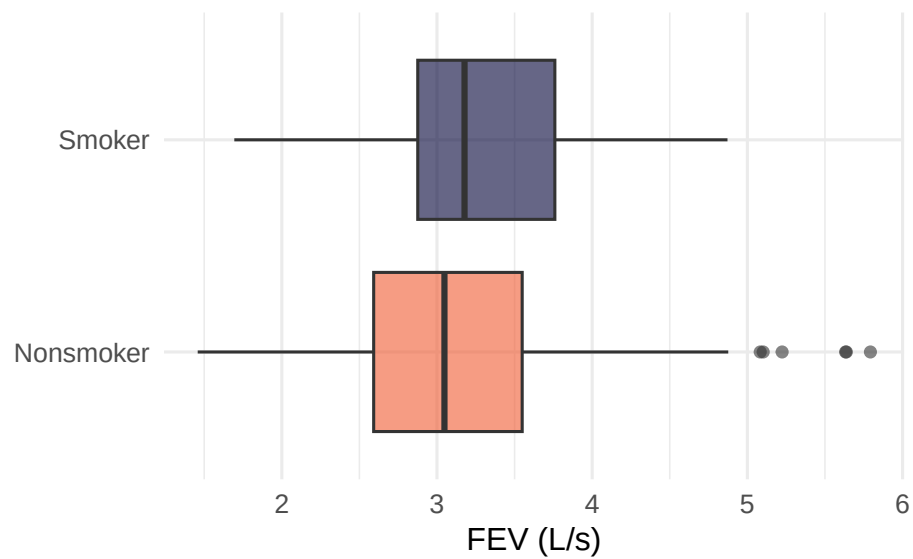
Wrinkle: lung function naturally increases during adolescence, but smoking may slow its progression – so we need to account for *both* age and smoking status.

Variable	Description
FEV	forced expiratory volume (liters/second)
Age	age in years
Smoke	Smoker or Nonsmoker

Note

Source: Rosner (2006)

EDA



Notice smokers skew *older* – any raw FEV difference between smokers and nonsmokers is tangled up with age. This is exactly why we need a model that controls for both variables at once.

Indicator variables

Regression requires a **numeric** representation of every variable, so a categorical predictor gets converted to indicator (0/1) columns – e.g. Smoker = , Nonsmoker = .

💡 Group task

For each regression model on the handout, sketch the fitted model on the whiteboard. Each fitted model will have two lines: one for smokers, one for nonsmokers.

❗ Stop here

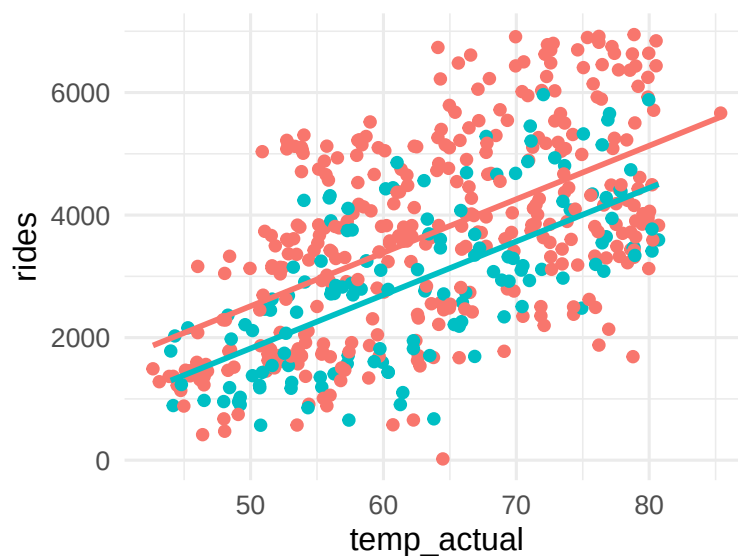
Let Amanda know when your group finishes!

Three ways to combine a line + a group

We'll illustrate with `rides ~ temp_actual + weekend` on the bikes data, since it's easy to see two clear groups.

Parallel lines model

Same rate of increase in both groups, but a different baseline.

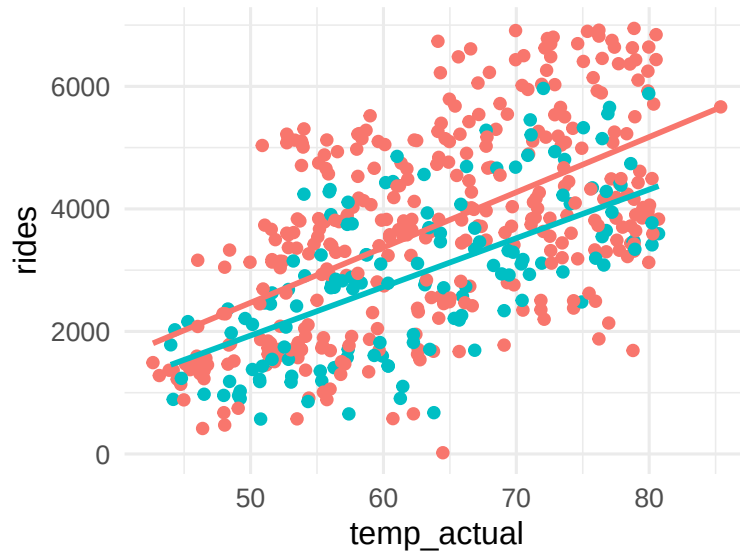


$$\mu(y|x) = \beta_0 + \beta_1 \text{temp} + \beta_2 \text{weekend}$$

- β_0 : intercept for weekdays
- $\beta_0 + \beta_2$: intercept for weekends
- β_1 : slope for _____

Different slopes model

Same baseline, but temperature affects ridership differently on weekends vs. weekdays.

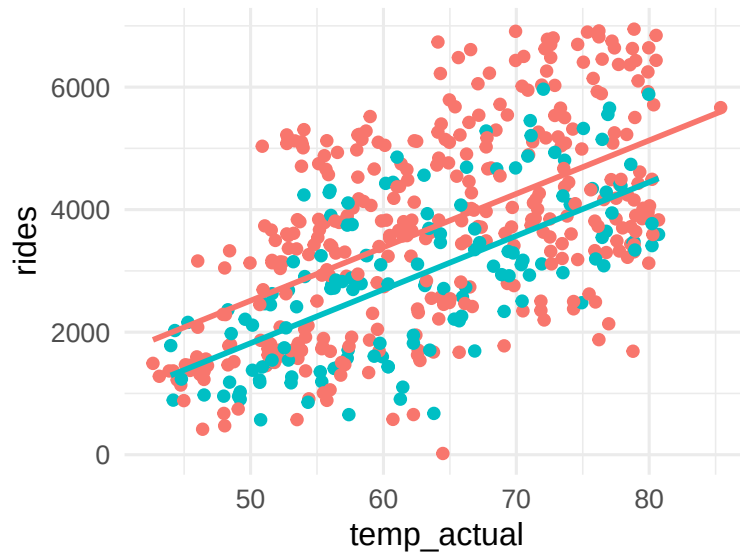


$$\mu(y|x) = \beta_0 + \beta_1 \text{temp} + \beta_2 \text{temp} \times \text{weekend}$$

- β_0 : intercept for _____
- β_1 : slope for weekdays
- $\beta_1 + \beta_2$: slope for weekends

Separate lines model

Both the baseline *and* the rate of increase differ by group.



$$\mu(y|x) = \beta_0 + \beta_1 \text{temp} + \beta_2 \text{weekend} + \beta_3 \text{temp} \times \text{weekend}$$

- β_0 : intercept for weekdays
- $\beta_0 + \beta_2$: intercept for weekends
- β_1 : slope for weekdays
- $\beta_1 + \beta_3$: slope for _____

i Note

A model with an **interaction term** ($x \times \text{group}$) is exactly the separate-lines model: it lets both the intercept and slope differ by group.

More than two categories

What if we want to incorporate season (fall, spring, summer, winter)?

A bad idea

Convert the column to a single number: Fall $\rightarrow$ 1, Winter $\rightarrow$ 2, Spring $\rightarrow$ 3, Summer $\rightarrow$ 4, then fit $\mu(\text{rides} \mid \text{season}) = \beta_0 + \beta_1 \text{season}$.

This is wrong – it forces the seasons onto an arbitrary numeric scale and assumes, e.g., that the “jump” from fall to winter is the same size and direction as the jump from spring to summer.

A good idea: indicator (dummy) variables

For a categorical variable with k levels, create $k - 1$ indicator variables (one level is left out as the **reference level**).

season	seasonwinter	seasonspring	seasonsummer
winter	1	0	0
fall	0	0	0
spring	0	1	0
spring	0	1	0
fall	0	0	0
summer	0	0	1

Here fall is the reference level – it's represented by all zeros – and each other season gets its own 0/1 column.

 Group task

For each sketched regression model on the handout, write the mean function that matches it (line type represents season).

 Stop here

Let Amanda know when your group finishes!